
name: spec description: Architectural specification drafting skill. Formulates rigorous, beginner-proof, and agent-executable system specifications based on clarified requirements.

Spec — Architectural Specification Generator

Purpose & Protocol

The **Spec** skill codifies architectural decisions into clear, permanent, and machine-readable specification documents. Specifications authored by this skill serve as the immutable ground truth for implementation agents and human operators.

Required Document Structure

Every specification must strictly adhere to this section layout:

1. **Document Identity & W5+H Subsystem Identity:**

- **Title & Version:** Canonical naming and semantic version.
- **WHO:** Which agent, daemon, or human persona executes or maintains this.
- **WHAT:** Exact capabilities, boundaries, and delivered modules.
- **WHEN:** Trigger events, lifecycle phases, cron schedules, or event hooks.
- **WHERE:** Physical paths, repository subfolder, socket address, port bindings.
- **WHY:** High-level rationale, technical trade-offs, and failure costs.
- **HOW:** Deterministic protocols, data schemas, CLI flags, and error codes.

2. **Core Architectural Invariants:**

- Numbered list of non-negotiable laws governing this module (e.g., "Invariant 1: Raw inputs are immutable and read-only").

3. **Data Schemas & Interface Contracts:**

- Full JSON schemas, Protobuf definitions, TypeScript interfaces, or Python dataclasses. No hand-waving or pseudocode.

4. Topology & Dataflow Diagram:

- ASCII / Unicode flow diagram illustrating ingress, transformations, and egress paths.

5. Failure Modes & Recovery Procedures:

- Specific exit codes, quarantine procedures, and automated rollback policies.

6. Acceptance Criteria & Verifiable Checklist:

- Concrete, observable conditions required for an auditor agent to certify completion.